VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Executive Summary and Document Catalog

AFLCMC/WA

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1.0 Executive Summary

1.1 The Challenge

In the constantly-evolving threat landscape of the 21st century, the United State's Warfighters need to meet and outpace each threat. In order to maintain tactical and technological advantage, it is critical that the United States of America (USA) maintains the ability to adopt new technologies at a faster pace than ever before. Standardization of protocols and interfaces supports this goal.

The unique challenges around new technology integration and the standardization of platform interfaces have an immediate impact on the United States Government (USG) and the warfighter, including:

- Vendor lock adds significant costs by impeding competition during procurement
- Reuse of existing software and/or hardware often requires custom adapters or reintegration
- Warfighters struggle to share applications at the speed of innovation

As the threat landscape grows through increasing frequency and sophistication of attacks, the use of proprietary solutions must diminish and a hybrid environment must emerge. The solution which benefits both the warfighter and the future of the USG begins with a non-proprietary approach to system architectures.

1.2 The Response - An Open Mission System Architecture

The solution to the identified challenges is an Open Mission Systems Reference Architecture that allows vendors to rapidly provide highly interoperable hardware and software capabilities. An architecture such as this creates the fundamental organization of Mission System Components, such as those identified in **Section 2**, as well as the architecture's guiding design principles[1] to its adopters.

Reference architectures tailor the taxonomy, terminology, and hardware artifacts for a specific domain. For example, an airline reference architecture may be focused on passenger metrics, hub management artifacts, and Federal Aviation Administration (FAA) policies.

Agile Mission Suite (AMS) Government Reference Architecture (GRA) is a reference architecture that specifically supports the unique requirements of Department of Defense (DoD) platforms. The AMS GRA includes policies, rules, interface descriptions, Starter Kits (SKs), Development Kits (DKs), and tools directly geared and applicable to Mission Systems. The development environments fully support and leverage all the benefits of agile software

[1] IEEE 1471, 2000.

development and Model-Based Systems Engineering (MBSE). The MBSE environments include Development, Security, and Operations (DevSecOps), Emulators/Simulation, Automated Test, and enterprise-level repositories for sharing data across stakeholders.

1.3 Agile Mission Suite (AMS) Government Reference Architecture (GRA)

AMS GRA is a USG-owned, digital MBSE product that specifies a Mission System architecture for weapon systems. AMS GRA represents a fundamental paradigm shift in balancing mission capabilities against the cost of acquisition, deployment, sustainment, and maintainability. Advances in Commercial-Off-The-Shelf (COTS) processing capabilities, high-speed network solutions, and digital Multi-Function Apertures (MFAs) enable concepts such as independent procurement of traditional subsystem components, reconfiguration of subsystems, and distributed processing for shared Virtual Apertures (VAs). AMS GRA is purpose-built to enable software-definable digital apertures to dominate the electromagnetic spectrum. AMS GRA defines a common taxonomy and common interfaces to enable the rapid adoption and utilization of state-of-the-art MFA and Multi-Function Processor (MFP) technologies. These capabilities enable the development and fielding of weapon systems that are inherently upgradeable with a significantly reduced development timeline and effort. AMS GRA allows upgrades to outpace the threat evolution cycle by embracing the key tenets of competition, interchangeability, reliability, and maintainability. AMS GRA fully incorporates and supersedes the Open Communications Subsystem (OCS) standard 2.0.

The open nature of AMS GRA drives innovation and continuous competition. This capability allows the USG to avoid vendor lock and reduce cost. AMS GRA achieves these objectives through the development of Minimum Procurable Units (MPUs) with Open and Common interfaces and well-defined behaviors.

1.3.1 Acquisition Enabled by the Architecture

AMS GRA is an innovative architectural and business approach enabling separate Mission System suppliers to efficiently use Common and Open architectural elements. AMS GRA uses the business and technical objectives included below to be agile, proactive, innovative, competitive, transportable, and affordable.

- **Agile**
  - Uses common, standardized interfaces to enable faster acquisition of new capabilities and reduce duplicative investments
  - Facilitates reuse of components and systems from multiple sources
  - Partitions functions that require different acquisition timelines
- **Proactive**
  - Permits the addition of new capabilities without hardware changes in response to current and emerging threats through the use of software-definable MFAs
  - Facilitates integration of new software-defined capabilities through the use of high-speed data distribution
  - Delivers improved software capabilities with minimal reintegration using loosely-coupled software architecture based on the DoD-approved DevSecOps methodology
- **Innovative**
  - Supports hosting Critical Program Information (CPI) and Multi-Level Security (MLS) across the architecture
  - Incubates new technologies in Government labs, small businesses, academia, and adjacent markets, and inserts these technologies without reworking existing capabilities or requiring extensive regression testing
  - Combines the high performance of MFAs with software-defined Skills
  - Enables sharing MFA resources across multiple operations
- **Competitive**
  - Avoids vendor lock by utilizing Open and Common interfaces, the provision of DKs, creation of new AMS GRA Services, and the extension and adaptation of the Open Mission Systems (OMS) software decomposition pattern to sensor subsystems
  - Enables a fair marketplace for industry competition of hardware and software components at all layers of the Mission System
  - Leverages a common DevSecOps environment for collaboration and rapid insertion of third-party software upgrades to outpace the threat
- **Transportable**
  - Facilitates adoption and maturation of partitioned architecture, established MPUs, and defined interfaces
  - Develops hardware and software that are platform- and vendor-agnostic through the use of Open or Common interfaces that incorporate commercial standards and commodity hardware
  - Enables new Skills or Skill upgrades without hardware upgrades or extensive retest, reverification, and revalidation of other components

- **Affordable**
  - Reduces initial acquisition and sustainment costs by relying on transportable features and the efficiencies of a common DevSecOps environment
  - Reduces integration and software development costs through effective tool-chains and modeling from hardware design to sustainment
  - Enables continuous competition throughout system lifecycles by utilizing partitioned architecture, established MPUs, and defined interfaces

AMS GRA enforces these attributes through compliance. Compliance is critical to the AMS GRA because it enforces adherence to interface definitions and documentation requirements. The AMS GRA uses a tiered approach for compliance that enables the application of AMS GRA to new designs and existing platforms. The Compliance Assessment Approach Volume describes the compliance strategy. Likewise, the complete, authoritative set of AMS GRA requirements for the Mission System, DKs, and all MPUs are captured in the Verification Cross Reference Matrix (VCRM) located in the Compliance Assessment Approach Volume. For reference, these requirements are also included in the respective MPU Volumes.

Figure 1.3.1 summarizes the principal tenets AMS GRA supports. AMS GRA achieves these Mission System quality attributes by requiring the following guiding principles:

- Standards and reference architecture developed and approved by the DoD/Industry consortium
- Government ownership of standards and appropriate Intellectual Property rights
- Independent compliance testing by Procuring Authorities, Mission System Integrators (MSIs), and innovators

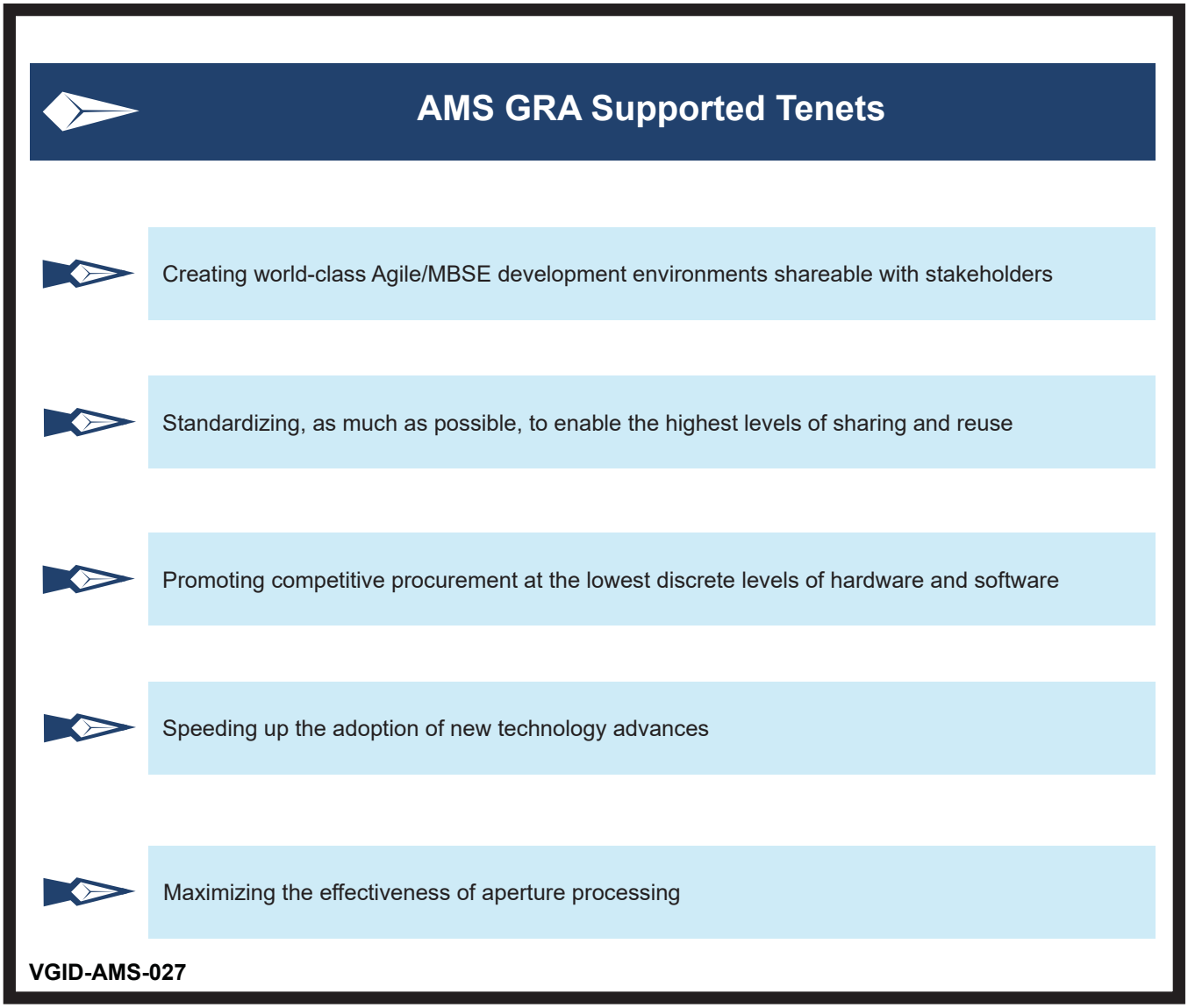


Figure 1.3.1: Alignment with the Five AMS GRA Supported Tenets Allows Adopting Platforms to Efficiently Meet the Mission Requirements.

1.3.2 Acceleration of New Operational Capabilities

AMS GRA establishes a foundation for a Mission System to accelerate the fielding of operational capabilities for weapon systems. The AMS GRA is a reusable technology architecture that can be adopted by a diverse set of acquisition programs. AMS GRA delivers even more value to the DoD by providing many capabilities such as rapid insertion of new technologies, embedded cybersecurity, superior individual and integrated performance of new technologies, and a technology-compatible acquisition environment.

- **Rapid Insertion of New Technology to Outpace Evolving Threats**
  - AMS GRA defines common interfaces for hardware and software. These interfaces enable low-risk technology insertion, ensuring the time line from development to fielding of capabilities is ahead of the adversary.
- **Embedded Cybersecurity that Enables Other Mission Needs**
  - As the USA and its warfighters encounter sophisticated cyber adversaries, AMS GRA's foundational cyber implementation ensures a resilient and enduring system.
- **Superior Individual and Integrated Performance of New Technologies**
  - The AMS GRA design facilitates utilization of shared MFAs through open architecture and open interfaces. This agile and easily adaptable approach supports software portability and enables innovation, which leads to unique ways to reach higher levels of integrated performance.
- **An Acquisition Environment Matched to the Enabling Technology**
  - The AMS GRA digital ecosystem provides a Government Purpose Rights (GPR) model-based SK. This SK provides a reference implementation of the AMS GRA platform and supporting tools to accelerate software development, integration, and deployment into an AMS GRA environment. An AMS GRA software factory that compiles the architecture, model-based documentation, and software framework allows incremental acquisition and shorter integration and development time scales. A description of the SK and its contents can be found in the AMS GRA SK overview document provided with the SK.

The AMS GRA supports multiple use cases for adoption. These use cases range from a complete Mission System all the way down to a partial hardware or software upgrade. The architectural elements defined by AMS GRA deliver reuse and acquisition benefits at all scales of adoption, with greater adoption yielding greater benefits for acquisition.

2.0 Technical Preface

The technical preface provides an introduction of the terminology, relationships, and functional concepts of AMS GRA. The preface also describes the top-level architecture and introduces the MPU classes.

2.1 AMS GRA Description

Figure 2.1.1 shows a high level breakdown of the AMS GRA and its various elements. AMS GRA partitions the architecture into three logically distinct and separate layers: the MFA, the MFP, and the Mission Processing Subsystem (MPS). The logical separation of MFA and MFP layers represents a decomposition of the traditional subsystem.

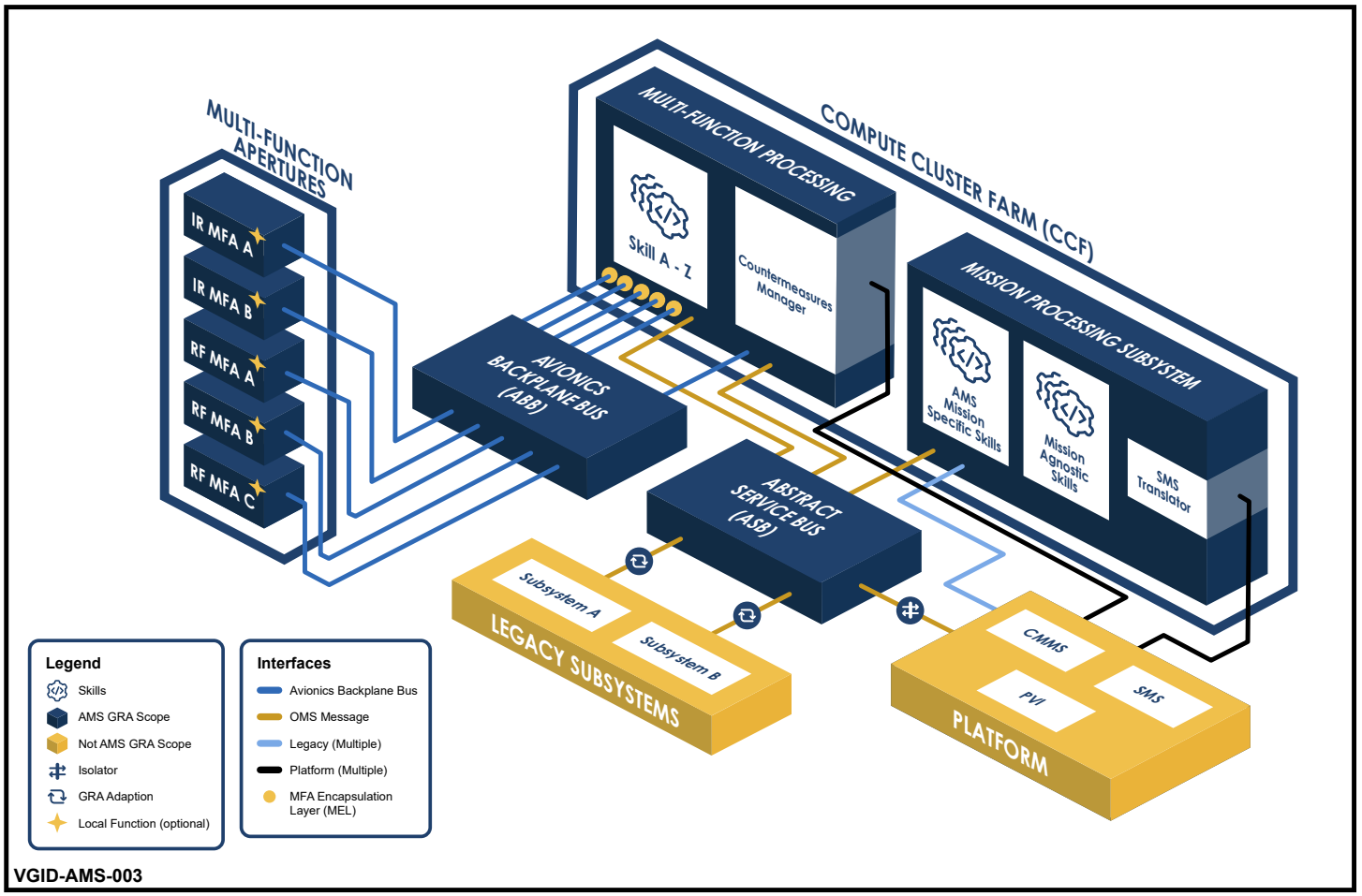


Figure 2.1.1: AMS GRA Encompasses a Wide Range of Infrastructure, Hardware, and Software Elements with Both Internal and External Connections.

2.1.1 MFA, MFP, and MPS Layers

The MFA layer includes a suite of software-defined, broadband, digital apertures spanning the Radio Frequency (RF) and Infrared (IR) spectrum. Unlike prior system architectures, these apertures do not solely implement any particular Skill or set of Skills. A key attribute of the MFA layer is MFA Local Functions with the provisioning to allow for insertion of new capabilities with third-party Local Functions.

The MFP layer contains a collection of processor types as defined by the Digital Payload Processor (DPP) MPU. These processors can be General Purpose Processors (GPPs), Field-Programmable Gate Arrays (FPGAs), or Graphics Processing Units (GPUs). The MFP Services represent the evolution of modes found in traditional sensing subsystems, decoupling the software skill from the compute hardware. In AMS GRA, capabilities are no longer stove-piped into single sub-systems or tightly-coupled to the sensing hardware. Traditional functional modes such as Air-to-Air radar search, radar warning, or direction-finding capabilities reside in both the MFP and MFA layers as Skills composed of Services deployed on DPP and Local Functions deployed on an MFA. In addition to general purpose computing and Skills, the MFP layer contains AMS GRA-defined services. These services implement Mission-specific capabilities which require access to the data on the Avionics Backplane Bus (ABB).

The MPS layer contains a collection of processing resources which provide a general processing capability. This layer is where functions such as System Track Management, Resource Management, and System Health reside. The set of available Level 3 and Level 2 MPUs within the MFP and MPS layers are the same. The MPS also includes multi-ship interfaces (e.g., Mission Platform MPS to MPS exchange) and interfaces to the Aircraft Systems and Pilot Vehicle Interface (PVI).

2.1.2 ASB and ABB

AMS GRA includes two main bus elements: the Abstract Service Bus (ASB) and the ABB. These architectural elements are logical constructs that are employed in the architecture and separated to maximize performance and enhance sensor resource sharing.

The ASB, defined in the OMS standards, is a composite architectural bus element that encompasses all forms of OMS Data Exchanges. AMS GRA leverages all of the standardized information exchanges defined in OMS to accelerate the adoption of new technology within the architecture.

The ABB facilitates the multi-functionality of the software-definable MFAs. The ABB is a high-speed network that allows the transport of minimally-processed aperture data to any of the DPPs within the MFP layer for further processing. This allows the flow of information from an MFA to multiple DPP consumers to support multiple functions, which breaks the traditional model of a single function tied to a single aperture. The ABB can also facilitate transport of countermeasure techniques generated by MFP or MFA layer MPUs to a Countermeasures Manager (CMM).

AMS GRA inherits OMS Data Transfer, OMS Messages (including Critical Abstraction Layer message schema), and OMS Special Signals. Each defined MPU has specific context and may include only a subset of these OMS communications.

AMS GRA implements a loosely-coupled Resource Management capability across the layers. The goal of the layered approach is to allow Resource Management functions to operate in the logical layer that makes the most sense for the specific Mission System instantiation. This layer specific implementation is done in alignment with overall Platform guidance. The Platform Resource Management sets informational needs and priorities based on mission objectives. The loosely-coupled Resource Management speeds local decision making by providing the layer specific implementation the autonomy to make decisions closer to the problem. Additionally, platforms can enhance the performance of the overall Resource Management capability by inserting higher performing resource management software into the specific layers without disrupting other architecture elements or resource management capabilities implemented in a different layer.

2.1.3 Security Framework and Interfaces

AMS GRA defines a Security Framework as a compilation of standard security interfaces, required Mission System characteristics, and MPU security compliance. The AMS GRA Security Framework provides an assessment of the following security principles:

- Cybersecurity
- Cyber Resiliency
- MLS
- Anti-Tamper (AT)

AMS GRA defines interfaces in order to support the communication and protection of security-relevant information. The Mission System may implement cryptographic mechanisms to support the bulk encryption of stored Data at Rest (DaR) as data.

AMS GRA accommodates the inclusion of legacy subsystems through adaptation. This adaptation may occur in hardware or software and includes the provision for special signals, including discrete signals, between Mission System elements. The adaptation enables the subsystems to be AMS GRA-compliant.

In any implementation of AMS GRA, some AMS GRA elements are expected to interact with non-AMS GRA elements (e.g., future non-AMS GRA systems and platform support services). Interactions which cross the AMS GRA boundary include interactions of the AMS GRA elements with other systems such as aircraft systems and PVI. When an interaction crosses the boundary, the Mission System may leverage adaptation and isolation patterns to ensure architectural updates do not trigger platform recertification.

2.2 AMS GRA Elements

AMS GRA defines the concept of Level 1, Level 2, and Level 3 MPU elements of the architecture as shown in **Figure 2.2.1**. MPUs are Mission System elements at the lowest procurable level of element decomposition. These MPUs support rapid upgrades and independent competition. An AMS GRA-compliant Mission System consists of a collection of MPUs and platform-provided facilities (e.g. communications cabling). While AMS GRA defines MPU classes, it does not enumerate the specific collection of MPU instantiations which compose a Mission System. Those decisions reside within the design process of the adopting program. Further information regarding the different MPUs can be found in their respective Volumes.

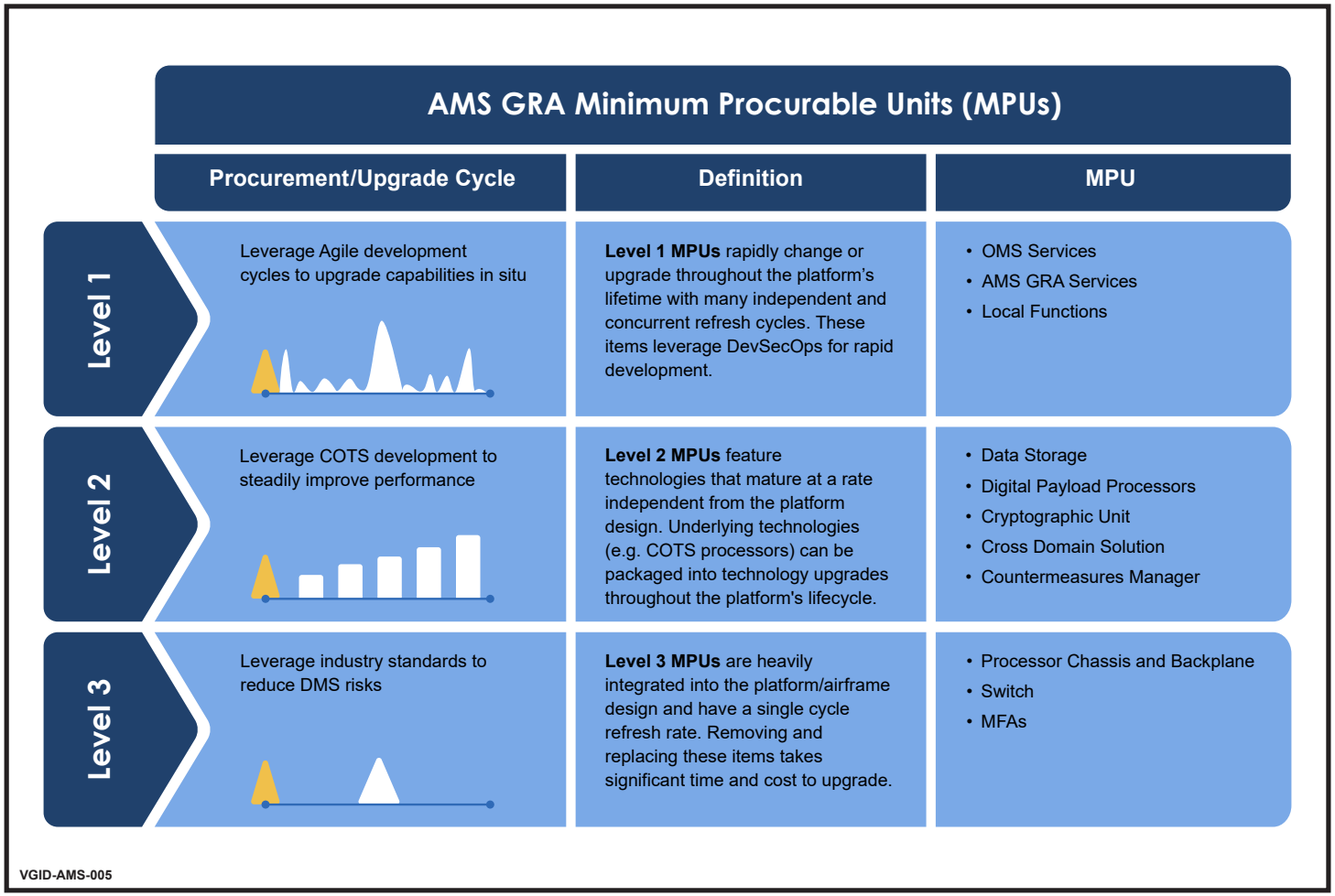


Figure 2.2.1: The AMS GRA's Three Levels of MPUs Allow Platforms to Identify, Plan, and Integrate New Capabilities in Alignment with Mission, Industry, and Certification Timelines.

Level 1 MPUs mature at the fastest rate and change in a rapid and continuous manner to implement new capabilities. Level 1 MPUs consist of AMS GRA Services, OMS Services, and MFA Local Functions as described in **Table 2.2.1**.

Level 1 MPU	Description
AMS GRA Service	Provides additional flexibility for ABB-related processing functions. Provides access to shared MPU hardware (e.g., MFA hardware by way of the ABB).
MFA Local Function	Provides computing for the MFA layer and enables low latency processing of data generated by an MFA MPU (e.g., timing, data transfer rates, and reduced control messaging).
OMS Service	Provides a standardized approach for software service development using the OMS standard for Mission System software.

Table 2.2.1: The Three Types of Level 1 MPUs Mature at the Fastest Rate and Change in a Rapid and Continuous Manner to Implement New Capabilities Throughout the Platform Lifecycle.

Level 2 MPUs mature at a slower rate than Level 1 MPUs and change as needed without a major platform refresh. AMS GRA prescribes Level 2 MPUs as itemized in **Table 2.2.2**.

Level 2 MPU	Description
CDS	Provides secure processing for CDS functions.
CMM	Provides non-safety-critical command and control via the ABB and/or the ASB.
Cryptographic Unit	Provides secure processing for cryptographic functions.
Data Storage	Provides general data storage, and may also provide encrypted and secure data storage accessible to specific AMS GRA elements, at the MSIs discretion, by way of the ASB, ABB, or both.
DPP	Provides the processing resources within a PCaB.

Table 2.2.2: The Five Types of Level 2 MPUs Mature Interdependently from the Platform Design and Can be Brought Into the Platform as Needed Without a Major Platform Refresh.

Level 3 MPUs mature at the slowest rate and only change as part of a major platform refresh. In general, these items are enduring parts of the platform. AMS GRA prescribes Level 3 MPUs itemized in **Table 2.2.3**.

Level 3 MPU	Description
MFA	Enables Skills to interact with the environment across frequency spectrum.
PCaB	Serves as the housing for and the interface between the platform and other MPUs including DPPs and possibly Cryptographic Units and CDSs, Switches, and Data Storage MPUs.
Switch	Provides the interface and routing for the switched network fabric and is a key enabler in implementing the ABB and ASB.

Table 2.2.3: The Three Types of Level 3 MPUs are Tightly Coupled to Platform Design and Require a Platform Refresh to Change.

AMS GRA introduces the concept of Skills to facilitate procurable software. Skills are collections of Level 1 MPUs that may exist in multiple layers. The Level 1 MPUs exist at specific layers of the architecture with loose coupling between them providing functional isolation. This functional isolation, along with the Open/Common interfaces and industry-provided DKs, allows new Skills to be independently developed and quickly integrated by either the Original Equipment Manufacturer (OEM) or an alternative Skill provider. See the Software Architecture Volume for additional information on Skills and DKs.

2.3 Mission Thread Example

Figure 2.3.1 shows an example Mission Thread provided to illustrate one aspect of how AMS GRA supports the needs of modern DoD Mission Platforms. Within this thread, the Emanation to Information (E2I) loop is integral to completing assigned mission parameters. This loop includes the sensors, processors, switches, software, and processing boards needed to take raw environmental emanations and convert them into actionable information that the pilot, mission subsystems, or the platform infrastructure can use to make decisions and take appropriate action. The E2I loop provides the situational awareness necessary for mission success. This example is intended to be representative of what occurs in an E2I AMS GRA loop, but does not represent every facet of the complex Mission Thread. Specifically, this example represents only the reception and not the active emanation portion of the loop[1]. The elements shown in **Figure 2.3.1** are described in greater detail in the AMS GRA Document Catalog in **Table 3.0.1**.

[1] There are varying ways for the E2I loop to be accomplished within AMS GRA

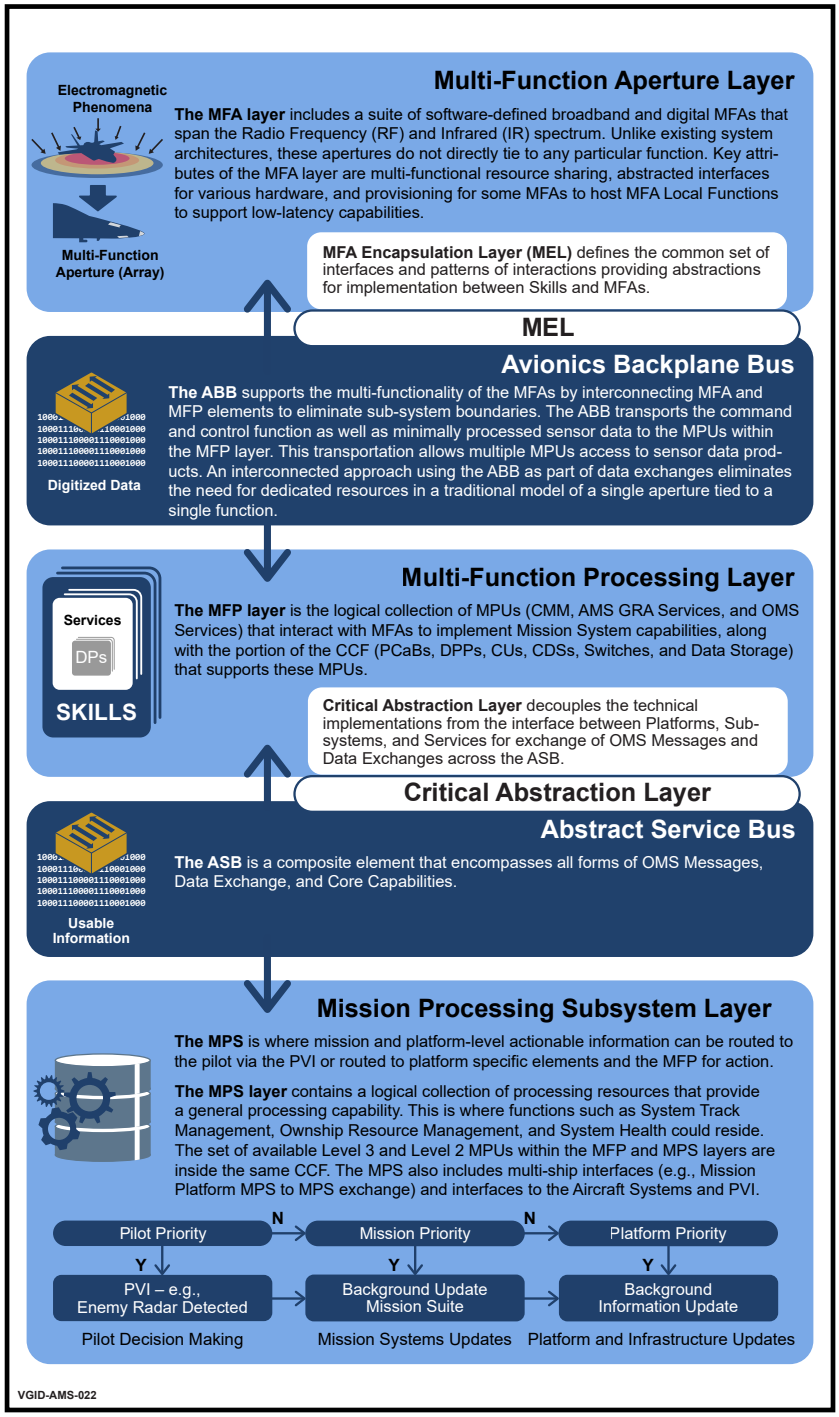


Figure 2.3.1: This E2I Mission Thread Emphasizes How the AMS GRA Supports Modern DoD Mission Platforms.

3.0 AMS GRA Document Catalog

The AMS GRA documentation is organized into a set of Volumes which allow the reader to focus on the most relevant content. Each Volume is intended to be the standalone authority for a specific topic or MPU; however, most users will need to reference multiple Volumes in order to understand how elements interoperate within an AMS GRA ecosystem. AMS GRA defines a set of user roles in relationship to procurement (referenced in **Table 3.0.1**). The following paragraphs provide several examples describing the relationship between user roles and the AMS GRA Volumes.

An MSI references the Mission System, Software Architecture, Security Framework, Resource Management, Mission Agnostic Service Infrastructure (MASI), Assembly, and Time Distribution Volumes to understand the overall Mission System requirements, integration patterns, and functional partitioning. Additionally, the MSI references applicable MPU, Interface Description Document (IDD), PVI, and Armament Volumes to understand requirements, expected behaviors, and interfaces for integration of MPUs and capabilities.

An AMS GRA Service MPU provider references the AMS GRA Service MPU, Software Architecture, Time Distribution, and MASI Volumes to understand MPU requirements, software architecture context, time interfaces, and execution environment. Additionally, the MPU provider may reference Mission System, Resource Management, IDD, or PVI Volumes for additional context and interface descriptions.

The AMS GRA Document Catalog in **Table 3.0.1** provides a brief description of each Volume, the user types most likely to find that Volume relevant, and the folder location for the Volume.

Volume	Description	User Type	Location
Executive Summary	Provides a high level description of AMS GRA, including technical goals and benefits of the architecture.	All	Top Level Folder
Glossary	Provides definitions of terms used throughout the AMS GRA documentation.	All	Top Level Folder
Compliance Assessment Approach	Describes and defines the tiered compliance approach of AMS GRA elements from individual MPUs to full Mission System. Additionally, the Compliance Assessment also describes the use of external standards as well as the relationship between compliance and an MPU developer, the MSI, and the Procuring Authority. Finally, the Compliance Assessment provides the full set of compliance requirements for AMS GRA.	All	Compliance
Mission System	Describes both the physical and architectural aspects of an AMS GRA-compliant Mission System, providing details on architectural considerations likely needed, but not wholly specified by AMS GRA. The Mission System volume provides guidance on how those architectural aspects can integrate into and are supported by the architecture, as well as the requirements that a Mission System must adhere to for AMS GRA compliance.	MSIs, Certification Authorities, MPU Providers	Architecture Volumes
Software Architecture	Defines and describes the software architecture elements used in AMS GRA, as well as the relationships between them.	MSIs, MPU Providers	Architecture Volumes
Resource Management	Describes how AMS GRA can be used to effectively manage the resources of the Mission System including Sensing and Processing elements.	MSIs	Architecture Volumes
Security Framework	Describes how the security-relevant framework is integrated throughout AMS GRA to achieve the necessary overall security architecture.	MSIs	Architecture Volumes
Mission Agnostic Service Infrastructure	Describes the functionality of the MASI.	MSIs, MPU Providers	Architecture Volumes
Armament	Describes an AIA that can be used by an MSI during the development of a Mission System. The Armament Volume also describes the SMS Common element which is required as part of a Mission System.	MSIs	Architecture Volumes
Time Distribution	Describes the time distribution options an MSI is allowed within an AMS GRA-compliant Mission System.	MSIs, MPU Providers	Architecture Volumes

Volume	Description	User Type	Location
Pilot Vehicle Interface	Describes the process an MSI should follow when developing a PVI to integrate with an AMS GRA-compliant Mission System.	MSIs	Architecture Volumes
Multi-Function Aperture MPU	Provides the logical context, interfaces, and key features of the MFA MPU, in addition to the requirements needed to assess the AMS GRA compliance of the MFA MPU.	MPU Providers, Certification Authorities	Architecture Volumes
Digital Payload Processor MPU	Provides the logical context, interfaces, and key features of the DPP MPU, in addition to the requirements needed to assess the AMS GRA compliance of the DPP MPU.	MPU Providers, Certification Authorities	Architecture Volumes
Processor Chassis and Backplane MPU	Provides the logical context, interfaces, and key features of the PCaB MPU, in addition to the requirements needed to assess the AMS GRA compliance of the PCaB MPU.	MPU Providers, Certification Authorities	Architecture Volumes
Switch MPU	Provides the logical context, interfaces, and key features of the Switch MPU, in additional to the requirements needed to assess the AMS GRA compliance of the Switch MPU.	MPU Providers, Certification Authorities	Architecture Volumes
Data Storage MPU	Provides the logical context, interfaces, and key features of the Data Storage MPU, in additional to the requirements needed to assess the AMS GRA compliance of the Data Storage MPU.	MPU Providers, Certification Authorities	Architecture Volumes
OMS Service MPU	Provides the logical context, interfaces, and key features of the OMS Service MPU, in addition to the requirements needed to assess the AMS GRA compliance of the OMS Service MPU.	MPU Providers, Certification Authorities	Architecture Volumes
AMS GRA Service MPU	Provides the logical context, interfaces, and key features of the AMS GRA Service MPU, in addition to the requirements needed to assess the AMS GRA compliance of the AMS GRA Service MPU.	MPU Providers, Certification Authorities	Architecture Volumes
Local Function MPU	Provides the logical context, interfaces, and key features of the Local Function MPU, in addition to the requirements needed to assess the AMS GRA compliance of the Local Function MPU.	MPU Providers, Certification Authorities	Architecture Volumes
Cryptographic Unit MPU	Provides the logical context, interfaces, and key features of the Cryptographic Unit, in addition to the requirements needed to assess the AMS GRA compliance of the Cryptographic Unit.	MPU Providers, Certification Authorities	Architecture Volumes
Cryptographic Unit IDD	Defines the common cryptographic functions and describes the Application Programming Interface (API) for the provided CU MPU interfaces for integration within a deployable platform	MPU Providers, Certification Authorities	IDDs

Volume	Description	User Type	Location
CDS MPU	Provides the logical context, interfaces, and key features of the CDS, in addition to the requirements needed to assess the AMS GRA compliance of the CDS.	MPU Providers, Certification Authorities	Architecture Volumes
CDS IDD	Describes the CDS functions and the common interface for communication with the CDS MPU.	MPU Providers, Certification Authorities	IDDs
CMM MPU	Provides the logical context, interfaces, and key features of the CMM MPU, in addition to the requirements needed to assess the AMS GRA compliance of the CMM MPU.	MPU Providers, Certification Authorities	Architecture Volumes
Cyber IDD	Describes the Cyber Resiliency Framework and provides a common interface for communication between Cyber Manager, Cyber Monitor, and Cyber Agent architectural elements.	MPU Providers	IDDs
CMM IDD	Describes CMM interfaces using both the ASB and ABB.	MPU Providers	IDDs
Common RF MEL IDD	Provides a common interface for services to communicate with RF MFAs.	MPU Providers	IDDs
Common IR MEL IDD	Provides a common interface for services to communicate with IR MFAs.	MPU Providers	IDDs
Assembly Volume	Provides guidance to a Mission System Integrator on how to best procure and integrate new or upgraded capabilities into AMS GRA-compliant Assemblies and Mission Systems. The volume also provides compliance requirements against which an AMS GRA-compliant Assembly is assessed.	MSIs, Certification Authorities	Architecture Volumes

Table 3.0.1: The AMS GRA Provides Readers a Set of Volumes. Each Volume is the Standalone Authority on that Section of the Architecture.

4.0 Change Log

This list is intended to provide a high-level overview of architectural, interface, and software product changes since the last SK release.

4.1 Architectural Additions and Updates

In version 14.0, the AMS GRA documentation has been updated to reflect the modernized Cryptographic Unit (CU) and Cross Domain Solution (CDS) MPUs. Additionally, the AMS Façade concept has been replaced by the Portable Operating System Interface (POSIX), promoting clarity for adopters while still maintaining design for modularity (see updated GRA-MPU-003). The AMS and OMS Service MPU Volumes introduce the new Standardized Marketplace where adopters can pull Services suited for their specific needs.

Multiple updates to MASI were made as a result of both architectural maturation and the integration with Big Iron (BI). The MASI Management Interface is introduced in the MASI Volume, aimed to satisfy common health and status reporting data flows for multiple MPUs. This includes the MASI Management API, which standardizes health and status functional reporting. Further clarification of MASI requirements, GRA-MASI-021 and GRA-MASI-22, is provided in the Software Architecture Volume regarding the approach to MPU software containerization.

Following the integration of Big Iron (BI) into AMS, the concept of an Assembly and corresponding documentation is added to provide procuring authorities flexibility in their Mission System procurement strategies.

4.2 Requirement Change Log

The requirements listed below reflect the architectural changes to AMS GRA version 14.0 identified in the section above. Additionally, a number of requirements were modified for clarity.

Table 4.2.1 summarizes changes to the requirements.

Id	Text	Revision Status	Date	Major Baseline
GRA-MASI-027	A MASI shall register the services available for deployment.	New	11/13/2025	SK 14.0
GRA-MASI-028	A MASI shall report the state of MASI	New	11/13/2025	SK 14.0
GRA-MASI-029	A MASI shall support deployment of a minimum of two instances of the same Service.	New	11/13/2025	SK 14.0
GRA-MASI-030	A MASI shall isolate AMS GRA and OMS Service software faults from other software MPUs.	New	11/13/2025	SK 14.0
GRA-MASI-031	A MASI shall assign an MPUs' deployment targets and resources in accordance with that MPUs's delivered service.yml file.	New	11/13/2025	SK 14.0
GRA-MASI-032	A MASI shall provide an implementation of the AMS Common Transport API.	New	12/2/2025	SK 14.0
GRA-MASI-033	A MASI shall report the status of all SW MPUs available in the system in UCI format.	New	12/17/2025	SK 14.0
GRA-MASI-034	A MASI shall report compute resource information in UCI format if requested in UCI format by an MPU within the system.	New	12/17/2025	SK 14.0
GRA-MPU-088	An MPU shall report health and status on its Management Interface.	New	11/11/2025	SK 14.0
GRA-MPU-089	An MPU shall implement interfaces according to the Cryptographic Unit IDD.	New	11/11/2025	SK 14.0
GRA-MPU-090	An MPU shall provide a configuration file named service.yml with contents in accordance with the Service MPU Configuration Template YAML file.	New	11/13/2025	SK 14.0
GRA-MPU-091	An MPU shall report BIT information on its Management Interface.	New	12/23/25	SK 14.0
GRA-MPU-092	An MPU shall include a software bill of materials in an open and non-proprietary format.	New	12/18/25	SK 14.0
GRA-MS-076	A Mission System shall instantiate the Cyber Agent.	New	12/08/25	SK 14.0
GRA-MS-077	A Mission System shall instantiate the Cyber Monitor.	New	12/08/25	SK 14.0
GRA-MS-078	A Mission System shall instantiate the Cyber Manager.	New	12/08/25	SK 14.0
GRA-MS-079	A Mission System shall publish the Service health and status messages on the ABB to the ASB	New	06/05/25	SK 14.0
GRA-MASI-003	A MASI shall monitor the Management Interface for reported health and status of AMS GRA-compliant hardware.	Modified	11/11/2025	SK 14.0
GRA-MASI-006	A MASI shall monitor fault information reported by AMS GRA-compliant MPUs.	Modified	11/11/2025	SK 14.0
GRA-MASI-008	A MASI shall consume all available BIT information reported by AMS GRA-compliant hardware.	Modified	12/23/25	SK 14.0
GRA-MASI-012	A MASI shall be able to respond to fault information reported by AMS GRA-compliant MPUs.	Modified	11/11/2025	SK 14.0

Id	Text	Revision Status	Date	Major Baseline
GRA-MASI-015	A MASI shall monitor the health and status reporting of all deployed AMS GRA and OMS Services.	Modified	12/08/2025	SK 14.0
GRA-MASI-016	A MASI shall monitor compute resource information (e.g., utilization, availability, capability, bandwidth, flash, RAM).	Modified	11/11/2025	SK 14.0
GRA-MASI-022	A MASI shall deploy OCI-compliant containerized software utilizing a container framework.	Modified	12/08/2025	SK 14.0
GRA-MASI-023	A MASI shall provide access to compute resource information.	Modified	11/11/2025	SK 14.0
GRA-MPU-003	An MPU targeting deployment on a GPP DPP shall be restricted to using the POSIX API for OS system calls.	Modified	11/11/2025	SK 14.0
GRA-MPU-004	An MPU shall meet all of the OMS Tier 1 requirements for an OMS Service.	Modified	11/11/2025	SK 14.0
GRA-MPU-018	An MPU shall implement interfaces according to the CDS IDD.	Modified	11/11/2025	SK 14.0
GRA-MPU-051	An MPU shall include a software bill of materials in an open and non-proprietary format.	Modified	12/18/25	SK 14.0
GRA-MPU-057	An MPU's DP that targets a GPP Processing Component shall consist of a single OCI-compliant container.	Modified	12/18/25	SK 14.0
GRA-MPU-071	An MPU shall be built, packaged, and delivered to persistent storage via automated pipeline.	Modified	06/05/25	SK 14.0
GRA-MS-006	A Mission System that includes SMS shall implement the WSI as defined in the Armament Volume.	Modified	11/11/2025	SK 14.0
GRA-MS-010	A Mission System shall provide an AMS GRA-compliant MASI implementation.	Modified	11/11/2025	SK 14.0
GRA-MS-023	A Mission System shall make available all MEL Implementations supplied by MFA providers to any AMS GRA Service to be loaded.	Modified	12/08/2025	SK 14.0
GRA-MS-033	A Mission System with classified data shall encrypt all classified DaR using approved cryptographic algorithms.	Modified	12/08/2025	SK 14.0
GRA-MS-059	A Mission System shall verify the integrity of a hardware MPU over a secure communications channel prior to enabling the transfer of software, mission data, or associated Key Material to/from that MPU.	Modified	11/11/2025	SK 14.0
GRA-MS-075	A Mission System shall provide a Critical Program Information (CPI) analysis, to include Consequence of Compromise (Consequence of Compromise (CofC)), in accordance with the Department of Defense Anti-Tamper Engineering Framework.	Modified	12/11/25	SK 14.0
GRA-MASI-005	RESERVED	Deleted	11/11/2025	SK 14.0
GRA-MASI-014	RESERVED	Deleted	06/05/25	SK 14.0
GRA-MASI-018	RESERVED	Deleted	12/08/2025	SK 14.0

Id	Text	Revision Status	Date	Major Baseline
GRA-MASI-019	RESERVED	Deleted	12/08/2025	SK 14.0
GRA-MASI-026	RESERVED	Deleted	12/08/2025	SK 14.0
GRA-MPU-016	RESERVED	Deleted	11/11/2025	SK 14.0

Table 4.2.1: The AMS GRA Provides a List of All Requirements that Have Changed Since the Last Major Release, Allowing Adopters to Understand the Compliance Impact on Their System Against the Architecture.

4.3 Interface Additions and Updates

- IR MEL:
  - Changed:
    - ChannelCapability.h : class ChannelCapability
    - FrameHeader.h : class FrameHeader
    - IRSTScanTask.h : class IRSTScanTask
    - IRSTStareTask.h : class IRSTStareTask
    - IRSTTrackTask.h : class IRSTTrackTask
  - Added:
    - NavError.h : class NavError
    - SensorNavState.h : class SensorNavState, enum class CoordinateSystemType
    - API_Manager.h and .cpp : class API_Manager
    - Buffer.h and .cpp : class Buffer
    - Channel.h and .cpp : class Channel
    - Config.h and .cpp : class Config
    - Control.h and .cpp : class Control
- RF MEL:
  - Changed:
    - CommonModulations.h : class WaveformBuffer
    - jobDataFormat.h : ComplexI8VecPtr, ComplexI16VecPtr, JobTransmitVector
    - MELComplex.h : class MELComplex
- Common MEL:
  - Added:
    - BIT_Command.h : class BIT_Command

4.4 Software Additions and Updates

As part of version 14.0, the AMS GRA codebase includes several updates to the Starter Kit:

- AMS GRA Release v2026.01.00
  - Added in a Big Iron starterkit
  - Added in a CMSS starterkit
  - Rearchitected COMPASS compliance utilized in AMS GRA Pipelines
- Big Iron Starterkit
  - Transitioned to Next Generation modular format

- CMSS Starterkit
  - Transitioned to Next Generation modular format
- MASI Starterkit v2026.01.04
  - Added in adapters for Big Iron application sets
  - Updated starterkit to pull from a slug as opposed to the GitLab API ‘linktodownload‘
  - Removed ASB Data Management module, plan to re-introduce next release
  - Incorporated 3rd party updates to runtime infrastructure package
- Module Release Notes
  - Release Notes for all modules are available on DLE

A.0 Acronyms

ABB	Avionics Backplane Bus
AIA	Armament Integration Architecture
AMS	Agile Mission Suite
AMTI	Air Moving Target Indicator
API	Application Programming Interface
ASB	Abstract Service Bus
AT	Anti-Tamper
BI	Big Iron
BIT	Built-In Test
CDS	Cross Domain Solution
CMM	Countermeasures Manager
CofC	Consequence of Compromise
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CU	Cryptographic Unit
DaR	Data at Rest
DevSecOps	Development, Security, and Operations
DK	Development Kit
DoD	Department of Defense
DP	Digital Payload
DPP	Digital Payload Processor
E2I	Emanation to Information
ESM	Electronic Support Measures
FAA	Federal Aviation Administration
FPGA	Field-Programmable Gate Array
GPP	General Purpose Processor
GPR	Government Purpose Rights

GPU	Graphics Processing Unit
GRA	Government Reference Architecture
IDD	Interface Description Document
IR	Infrared
MASI	Mission Agnostic Service Infrastructure
MBSE	Model-Based Systems Engineering
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MLS	Multi-Level Security
MPS	Mission Processing Subsystem
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
OCI	Open Container Initiative
OCS	Open Communications Subsystem
OEM	Original Equipment Manufacturer
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
POSIX	Portable Operating System Interface
PTW	Protected Tactical Waveform
PVI	Pilot Vehicle Interface
RAM	Random Access Memory
RF	Radio Frequency
RWR	Radar Warning Receiver
SK	Starter Kit
SMS	Stores Management System
UCI	Universal Command and Control Interface
USA	United States of America
USG	United States Government

VA	Virtual Aperture
VCRM	Verification Cross Reference Matrix
WSI	Weapon Systems Interface
YAML	YAML Ain't Markup Language

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the SK continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	Chapter 4 of the document has been updated to reflect the new changes performed during Phase 13.	July 24, 2025	SK13.0
B	Chapter 4 of the document has been updated to document the changes implemented in version 14.0. Section 3 has been updated to include references to the CU IDD, the CDS IDD, and the Assembly Volume.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Executive Summary Volume Between Documentation Releases.